SI 140A-01 Probability & Statistics for EECS, Fall 2026
Homework 5

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Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1

If $X \sim \text{HGeom}(w, b, n)$, what is the distribution of $n - X$? Give a short proof.

Solution

Since $X \sim \text{HGeom}(w, b, n)$,

$$P(X = k) = \frac{\binom{w}{k}\binom{b}{n-k}}{\binom{w+b}{n}}.$$

Let $Y = n - X$. Then $X = n - Y$, so

$$P(Y = k) = P(X = n - k) = \frac{\binom{w}{n-k}\binom{b}{k}}{\binom{w+b}{n}}, \qquad \max(0, n - w) \leq k \leq \min(n, b).$$

Therefore,

$$n - X \sim \text{HGeom}(b, w, n).$$

Problem 2

A particular Youtuber is known for his arbitrary steak-eating habit during live streaming. In each live streaming day, he orders a steak with doneness as one of well done, medium well, medium, medium rare, and rare (with equal probability, independent of other live streaming days). How many days of live streaming do you expect to watch before you see him eating steaks with each possible doneness at least one once (suppose you are a big fan who watches every live streaming)?

Solution

After seeing $i-1$ distinct levels of doneness, let W_i be the number of additional days needed to see a new level, including the day on which it appears. The total number of days is

$$T = W_1 + W_2 + W_3 + W_4 + W_5.$$

For independent trials with success probability p, let W count the trials up to and including the first success. Then

$$P(W = t) = (1-p)^{t-1}p, \qquad t = 1, 2, 3, \ldots.$$

To find its expectation, write $q = 1 - p$. Differentiating the geometric series gives

$$\sum_{t=0}^{\infty} q^t = \frac{1}{1-q} \quad \Longrightarrow \quad \sum_{t=1}^{\infty} t q^{t-1} = \frac{1}{(1-q)^2}, \qquad 0 \le q < 1.$$

Hence,

$$\mathrm{E}[W] = \sum_{t=1}^{\infty} t(1-p)^{t-1}p = \frac{p}{(1-q)^2} = \frac{1}{p}.$$

For W_i, the probability of seeing a new level is

$$p_i = \frac{5-(i-1)}{5} = \frac{6-i}{5}, \qquad \mathrm{E}[W_i] = \frac{5}{6-i}, \qquad i = 1, \ldots, 5.$$

By linearity of expectation,

$$\begin{aligned} \mathrm{E}[T] &= \mathrm{E}[W_1] + \mathrm{E}[W_2] + \mathrm{E}[W_3] + \mathrm{E}[W_4] + \mathrm{E}[W_5] \\ &= 1 + \frac{5}{4} + \frac{5}{3} + \frac{5}{2} + 5 \\ &= 6 + \frac{15 + 20 + 30}{12} = \frac{137}{12} \approx 11.42 \text{ days.} \end{aligned}$$

Problem 3

Mario and Zelda are independently performing independent Bernoulli trials. For concreteness, assume that Mario is flipping a nickel with probability p_1 of Heads and Zelda is flipping a penny with probability p_2 of Heads. Let $X_1, X_2, \cdots$ be Mario's results and $Y_1, Y_2, \cdots$ be Zelda's results, with $X_i \sim \text{Bern}(p_1)$ and $Y_j \sim \text{Bern}(p_2)$.

(a) Find the distribution and expected value of the first time at which they are simultaneously successful, *i.e.*, the smallest n such that $X_n = Y_n = 1$.

(b) Find the expected time until at least one has a success (including the success).

(c) For $p_1 = p_2$, find the probability that their first successes are simultaneous, and use this to find the probability that Mario's first success precedes Zelda's.

Solution

(a) Let T be the first trial on which Mario and Zelda are simultaneously successful. By independence, the probability of simultaneous success in each trial is $p_1 p_2$. Thus,

$$P(T = n) = (1 - p_1 p_2)^{n-1} p_1 p_2, \qquad n = 1, 2, 3, \ldots.$$

So T has a geometric distribution on $\{1, 2, 3, \ldots\}$ with success probability $p_1 p_2$. Using the expectation derived in Problem 2,

$$\mathrm{E}[T] = \frac{1}{p_1 p_2}.$$

(b) The probability that both fail in a trial is

$$q = (1 - p_1)(1 - p_2).$$

Let S be the first trial on which at least one succeeds. Its success probability is $1 - q$, so

$$\mathrm{E}[S] = \frac{1}{1 - q} = \frac{1}{p_1 + p_2 - p_1 p_2}.$$

(c) Let $p_1 = p_2 = p$, and let M be the event that their first successes are simultaneous. For this to happen on trial n, both must fail in the first $n - 1$ trials and both succeed on trial n. Therefore,

$$\begin{aligned} P(M) &= \sum_{n=1}^{\infty} \left[(1-p)^2\right]^{n-1} p^2 \\ &= p^2 \left[1 + (1-p)^2 + (1-p)^4 + \cdots\right] \\ &= \frac{p^2}{1 - (1-p)^2} = \frac{p^2}{p(2-p)} = \frac{p}{2-p}. \end{aligned}$$

By symmetry, Mario and Zelda are equally likely to have the earlier first success. Let p' be the probability that Mario's first success precedes Zelda's. Then

$$2p' + P(M) = 1, \qquad p' = \frac{1}{2}\left(1 - \frac{p}{2-p}\right) = \frac{1-p}{2-p}.$$

Problem 4

Show that for any events $A_1, \ldots, A_n$,

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) \geq \sum_{j=1}^{n} P(A_j) - n + 1.$$

Hint: First prove a similar statement about indicator r.v.s, by interpreting what the events $I(A_1 \cap A_2 \cdots \cap A_n) = 1$ and $I(A_1 \cap A_2 \cdots \cap A_n) = 0$ mean.

Solution

For any event A, define its indicator by

$$I(A) = \begin{cases} 1, & \text{if } A \text{ occurs,} \\ 0, & \text{if } A \text{ does not occur.} \end{cases}$$

First, we prove that

$$I(A_1 \cap \cdots \cap A_n) \geq \sum_{j=1}^{n} I(A_j) - n + 1.$$

(i) If all the events occur, the left-hand side is 1 and the right-hand side is $n - n + 1 = 1$.

(ii) If at least one event does not occur, the left-hand side is 0 and $\sum_{j=1}^{n} I(A_j) \leq n - 1$. Thus, the right-hand side is at most 0, so the inequality also holds.

Taking expectations on both sides gives

$$\begin{aligned} \mathrm{E}[I(A_1 \cap \cdots \cap A_n)] &\geq \mathrm{E}\left[\sum_{j=1}^{n} I(A_j) - n + 1\right] \\ &= \sum_{j=1}^{n} \mathrm{E}[I(A_j)] - n + 1. \end{aligned}$$

Since $\mathrm{E}[I(A)] = P(A)$, we obtain

$$P(A_1 \cap \cdots \cap A_n) \geq \sum_{j=1}^{n} P(A_j) - n + 1.$$